Hong Kong Institute of Vocational Education
Discipline of Information Technology

HD in Data Science and Analytics (IT114116)

Final Year Project (ITP4870M)

Interim Report

The interim report must be submitted to Moodle by 7 February 2022, 23:55.

Please refer to the following guidelines for writing the Interim Report.

Guidelines for Writing the Interim Report

You are expected to look in sufficient detail at the project to be able to plan your work. This will involve preliminary work on the analysis of the problem and even some outlines of the designs. The structure and contents of the initial report may include, but not limited to:

Cover Page (see the sample given)

Abstract: brief description of the objectives of the project (about half page)

Table of Content (state page number clearly)

Table of Figures (optional)

1. Introduction

> ***Background and Motivation***
>
> ***Objectives and Scope***
>
> ***Outline of Report***

2. Problem Analysis and Proposed Solution

> ***Problem Analysis***
>
> ***Problem Statement***
>
> ***Existing Works***
>
> ***Proposed Solution***
>
> ***Scope and Limitation***

3. Data Analytical Framework

Data Acquisition and Understanding

Explain how data are acquired

Data Models / Algorithms

How data are processed and how data cleaning is done?

Explain why such approach is used, including the pros and cons.

Data Training and Testing

Performance Metrics

What are the criteria of measuring the results?

Data Presentation and Application

How data is visualized?

How the output will be fed back to the environment?

Probably, a software application, web interface and/or etc. will be created for user to interact, manipulate or make use of the output data.

Constraints and Limitations

For example: time constraint (to be finished in 8 months), availability of data source, data requirement, and other resources… etc.

4. Data Product Implementation

Overview of the Solution

Target Users / Expected Stakeholders of the solution

Functional and non-Functional Requirements

(A) Software Application (if final deliverable is a software system)

System Architecture

- Hardware, Software and Network Configuration for your system

System Design

- Software Architecture
- Use Case Diagram and Use Case Descriptions,
- Data Design (ERD and Data Dictionary)
- Other related design documents as appropriate

Interface Design

Hardware and Software Requirements

Such as software libraries, tools … etc.

Technical Considerations

Performance requirements…etc.

(B) Knowledge or Data Analytical Product

(For brainstorming)

Acquired Data Overview

- Assumptions
- Requirements
- Dictionary
- Original Dataset(s) sample

Data Preparation

- ETL approach
- Cleansing (optional, provide code if necessary)
- Manipulated Dataset(s) sample

Data Modelling

- Choose of Models
- Data Visualization Approach
- Predict outcomes

Data Presentation and Innovation

- Reporting Maturity Model
  - Dashboard design
  - Performance Indicators in Dashboard
- Innovative findings based on Reporting Model (For pure data project only)
  - Finding with statistical and supportive evidence

Business Impacts / Decision Making proposal

- Comparison of typical business concerns and Findings
- Suggestions
- Application Proposal for findings

5. Project Plan and Progress

Schedule (from Sept. 2021 – May 2022)

gantt chart specifically for your project

state milestone clearly

Milestone description

Progress

Difficulties Encountered

Work Distribution

Task and role of each project member

Action Plan for (the first Prototype)

Action Plan (what you are going to implement in the first prototype)

Feedback (what feedback you want to get from the prototype testing#)

References

Such as documents, websites, software libraries, models, audio…etc

Appendices

Design Thinking – Empathy

Describe the user experience on similar games in the market (both pros and cons)

Design Thinking – Define

What elements could make your game more enjoyable?
List the reframe the needs and insight

Design Thinking – Ideate*

Concentrate on idea generation (divergent thinking)
List all the ideas
Evaluate** different proposals (convergent thinking)
Select the best choice

\# Feedback from prototype testing helps to improve the product better, try to shorten the prototype development, the sooner the feedback obtained, the faster the project succeed.

* Students are encouraged to have **open-minded collaboration** to come up as many ideas as possible during the Ideate Stage.

** Do NOT ban difficult, yet innovative proposals too soon, **courage** from team members are essential, students are encouraged to carry on even if there are constraints or obstacles.

Prototype / Evaluation

Budget Estimation

Log Sheets (see the sample given) … etc.

(Some abandoned designs that you wanted to show may put here too)

You may need to adjust the above report structure to suit for your own need.

Because it is an early report and will affect the whole direction of the project, it is essential to get it right. For this reason, you must submit a draft *one week before the deadline*. Your supervisor will look through it and comment constructively in a general way on the draft, but will not give specific guidance or attempt to grade it. You should incorporate this advice into the final version of the report.

Important Notes:

- ***Group Leader*** is responsible to submit the Report on behalf of the group, i.e. only one submission is needed for each group
- You are required to ***hand-in one zip file named with your Group ID***
- The ***Directory Structure*** of the Zip file is:
    - Report (which may include also log sheets and appendix documents)
    - Software (the programs that you wrote (if any), models that you created, and libraries you used)